

Session Objectives

1. Students will be able to describe possible acute response to trauma
2. Students will be able identify groups at risk for a traumatic stress reaction.
3. Students will be able to appreciate initial and psychological First Aid interventions to a disaster situation.
4. Students will be able to describe the main symptoms clusters of PTSD.
5. Students will know the importance of screening for PTSD and what the options are for the initial management of patients with PTSD.

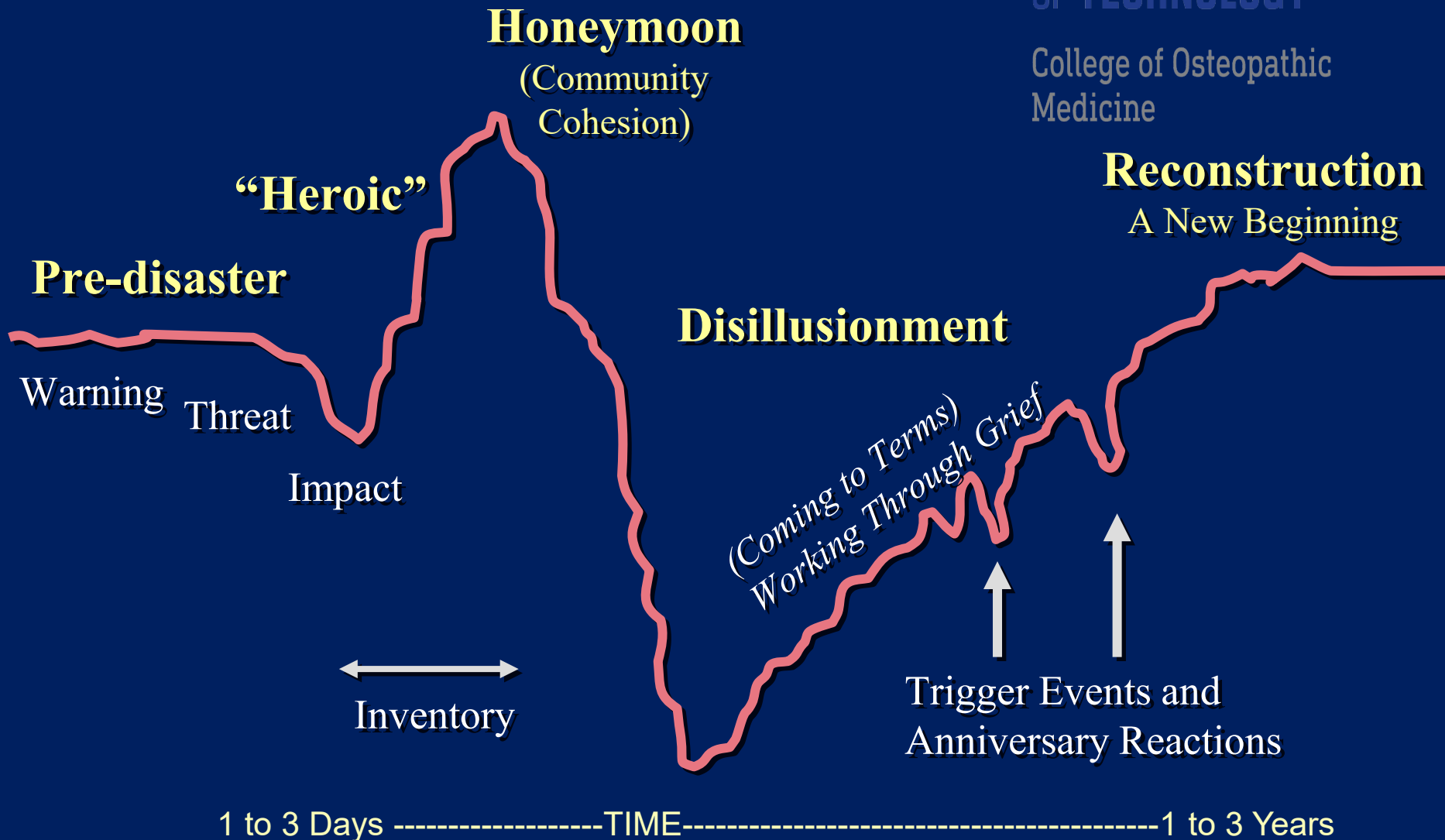
Mental Health Issues Coronavirus

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- Social Distancing ➡ boredom, loneliness, isolation, depression
- Quarantine ➡ insomnia, reduced feelings of safety, scapegoating, Alcohol/Tobacco
- Media ➡ amplifies Social Media-can misinform or be helpful
- Fear and uncertainty of the unknown
- Anxiety and guilt and stigma
- Protecting mental health of healthcare workers
- Telemental health, smart phone
- Anticipate and address and counsel about stress reaction
- Long COVID/Neuropsychiatric Sequelae

Source: Jeffrey Goldberg, DO



• Source: Zunin/Meyers

- **Key Concepts of Disaster Impact**

No one who sees a disaster is untouched by it.

- Primary, Secondary and Tertiary victims

Two types of trauma

- Individual (*stress and grief reactions*)
- Collective (*Damages the bonds and social fabric of the community. Increases fatigue, irritability, family conflict.*)

• Key Concepts of Disaster Impact

- People pull together during and after a disaster. (*high activity/low efficiency*)
- Stress and grief are normal reactions to an abnormal situation. (*transitory reactions*)
- Emotional reactions relate to problems of living. (*abnormal and excessive disruptions to daily routines*)

Key Concepts of Disaster Impact

- Disaster relief – can seem complex and overwhelming
- People typically do not seek out mental health services (*self reliance at all costs*)
- Survivors reject help (*others need it more than I do*)
- Mental health services are practical rather than psychological

• Critical Disaster Stressors

- Threat to one's life
- Threat of harm to family
- Destruction of home or community
- Significant media attention
- Witnessing others' trauma
- Being trapped or unable to evacuate
- Living with uncertainty

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Source: Adapted from CMHS

Cognitive Reactions to a Disaster

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- Trouble concentrating or remembering things
- Difficulty making decisions
- Preoccupation with the event
- Recurring dreams or nightmares
- Questioning spiritual beliefs

- **Physical Reactions to a Disaster**

Headaches

Generalized discomfort, hot or cold

Hypertension, heart pounding

Gastrointestinal distress

Exacerbation of psychiatric illness

Accelerated physical decline

Fatigue or exhaustion

Increase or decrease in appetite

- **Mental Disorders Secondary to Medical Condition**

- Head Injury
- Toxic Exposure
- Illness / Infection
- Dehydration
- Loss of Medications
- Withdrawal Symptoms

Comparison of Physical/Emotional Symptoms in Months Following 9/11

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<u>Symptoms</u>	NTS 2001		NTS 2002	
	National	New York	National	New York
Didn't feel like eating	28.8	46.4	15.0	16.2
Smoked more than usual	21.2	20.1	10.5	13.8
Had headaches	21.7	24.3	17.3	25.1
Had upset stomach	36.7	34.8	19.0	16.1
Cried	60.3	74.1	20.6	31.6
Had trouble getting to sleep	51.2	59.4	30.9	39.0
Felt very nervous or tense	49.9	62.5	26.9	33.6
Felt like getting drunk	6.6	11.9	8.6	10.7
Felt more tired than usual	37.5	47.6	32.5	41.2
Felt dizzy at times	9.0	13.3	10.9	19.8
Lost my temper more than usual	19.9	27.2	17.7	20.3
Hands sweated and felt damp and clammy	9.4	10.6	6.0	10.2
Had rapid heart beats	16.0	21.6	11.3	20.5
Felt sort of dazed and numb	45.7	46.3	12.3	17.3
Kept forgetting things	19.7	27.9	21.1	34.0
None of these	9.6	2.9	37.9	22.6

Source: University of Chicago, National Opinion Research Center, National Tragedy Survey 2001-2002

- **Factors Affecting Children's Recovery From a Disaster**

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Developmental level of child

Pre-disaster mental health of the child

Ability of the community to offer support

Separation from parents

Reaction of significant adults to the disaster



Older Adults Reactions to a Disaster: Stress Symptoms

- Slower to recover psychologically and financially
- Fear of loss of independence
- Depression
- Withdrawal
- Agitation
- Sleep disturbance
- Memory loss
- Disorientation and confusion
- Apathy

People with Mental Illness

- Same basic needs as everyone else
- May have special needs
- May “rise to the occasion”
- Program should tailor services to ensure appropriate services are delivered
- Often identify unserved or underserved people in completing outreach

People with Mental Illness

- Disaster stress reactions may be difficult to discern from symptoms of mental illness
- Consumers may be trained as part of the preparedness process
- Should be given the opportunity to serve the larger community

Disaster Workers

- Train peers when possible
- Recognize unique stressors
 - Conflicting roles (*family vs. job*)
 - Exposure to chaos, death and destruction
 - Very long days, exhaustion
 - Community reactions (*hero or scapegoat*)

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The Tampa Tribune / Michael Spooner

Stress Management for Healthcare Providers

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- Monitoring basic needs
- Clear communication
- Regular breaks
- Participation in normal life events
- Contact with loved ones
- Place for support
- Respect differences
- Preparedness

• Psychological First Aid

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Helping Victims in the Immediate Aftermath of Disaster

Do's:

- Promote Safety
- Promote Calm
- Promote Connectedness
- Promote Self-Efficacy
- Promote Hope

• Psychological First Aid

Helping Victims in the Immediate Aftermath of Disaster

Don'ts:

- Force people to share their stories with you, especially very personal details.
- Give simple reassurances like “everything will be OK” or “at least you survived.”
- Tell people what you think they should be feeling, thinking or doing now or how they should have acted earlier.

• Psychological First Aid

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Helping Victims in the Immediate Aftermath of Disaster

Don'ts (continued):

- Tell people why you think they have suffered by alluding to personal behaviors or beliefs of victims
- Make promises that may not be kept
- Criticize existing services or relief activities in front of people in need of these services

• Pharmacotherapy

Sleep Induction

Reduction of Anxiety

Acute Stress Disorder Symptoms

- Antidepressants?
- Propranolol?
- Risperidone?

No evidence of a PTSD protective benefit with benzodiazepines

Summary Statements

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Rescue workers and mental health teams can play a vital role in ameliorating the mental health problems of disaster survivors through:

- Awareness of risk factors
- Appropriate screening
- Initial Mental Health Intervention
- Psychological First Aid



Criterion A: Stressor

The person was exposed to: death, threatened death, actual or threatened serious injury, or actual or threatened sexual violence, as follows: (one required)

1. Direct exposure.
2. Witnessing, in person.
3. Indirectly, by learning that a close relative or close friend was exposed to trauma. If the event involved actual or threatened death, it must have been violent or accidental.
4. Repeated or extreme indirect exposure to aversive details of the event(s), usually in the course of professional duties (e.g., first responders, collecting body parts; professionals repeatedly exposed to details of child abuse). This does not include indirect non-professional exposure through electronic media, television, movies, or pictures.



Criterion B: Intrusive Symptoms

The traumatic event is persistently re-experienced in the following way(s): **(one required)**

1. Recurrent, involuntary, and intrusive memories. Note: Children older than six may express this symptom in repetitive play.
2. Traumatic nightmares. Note: Children may have frightening dreams without content related to the trauma(s).
3. Dissociative reactions (e.g., flashbacks) which may occur on a continuum from brief episodes to complete loss of consciousness. Note: Children may reenact the event in play.
4. Intense or prolonged distress after exposure to traumatic reminders.
5. Marked physiologic reactivity after exposure to trauma-related stimuli.



Criterion D: Negative Alterations in Cognitions and Mood

Negative alterations in cognitions and mood that began or worsened after the traumatic event: **(two required)**

1. Inability to recall key features of the traumatic event (usually dissociative amnesia; not due to head injury, alcohol, or drugs).
2. Persistent (and often distorted) negative beliefs and expectations about oneself or the world (e.g., "I am bad," "The world is completely dangerous").
3. Persistent distorted blame of self or others for causing the traumatic event or for resulting consequences.
4. Persistent negative trauma-related emotions (e.g., fear, horror, anger, guilt, or shame).
5. Markedly diminished interest in (pre-traumatic) significant activities.
6. Feeling alienated from others (e.g., detachment or estrangement).
7. Constricted affect: persistent inability to experience positive emotions.



Specify if: With Dissociative Symptoms.

In addition to meeting criteria for diagnosis, an individual experiences high levels of either of the following in reaction to trauma-related stimuli:

- 1. Depersonalization:** experience of being an outside observer of or detached from oneself (e.g., feeling as if "this is not happening to me").
- 2. Derealization:** experience of unreality, distance, or distortion (e.g., "things are not real" "this is more like a dream").



Specify if: With Delayed Expression.

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Full diagnosis is not met until at least six months after the trauma(s), although onset of symptoms may occur immediately.

Source: **DSM-5 APA (2013); PTSD: National Center for PTSD**



Specify if: **Preschool Subtype**

- Children 6 years or younger
 - Immediate reaction is not assessed. Children may not be able to express that they've been exposed to a traumatic event, if there were no adults around.
- Intrusive symptoms
 - Research has shown children do not always manifest with overt intrusive symptoms
- Avoidance and Negative alterations
 - Highly internalized phenomena
 - Only 1 symptom is required (instead of 3)
- Increased Arousal Symptoms
 - Most easily recognizable – extreme tantrums

Assessment Tools: PC-PTSD

Clarification

If the PC-PTSD screening instrument is utilized, clarify responses to determine:

- **Whether the patient has had a traumatic experience.**

"I notice from your answers to our questionnaire that you experience some symptoms of stress. At some point in their lives, many people have experienced extremely distressing events such as combat, physical or sexual assault, or a bad accident, and sometimes those events lead to the kinds of symptoms you have. Have you ever had any experiences like that?"

Assessment Tools: PC-PTSD

Clarification

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- **Whether reported symptoms are really trauma-related.**

"I see that you have said you have nightmares about or have thought about an upsetting experience when you did not want to. Can you give me an example of a nightmare or thinking about an upsetting experience when you didn't want to?" If a patient gives an example of a symptom that does not appear to be in response to a traumatic event (e.g., a response to a divorce rather than to a traumatic event), it may be that he or she is ruminating about a negative life event rather than experiencing intrusive thoughts about a traumatic stressor.

- **Whether reported symptoms are disruptive to the patient's life.**

"How have these thoughts, memories, or feelings affected your life? Have they interfered with your relationships? Your work? How about with recreation or your enjoyment of activities?"

Treatments

- Cognitive Behavioral Therapy (CBT)
 - Most well studied form of treatment
 - Helps to correct faulty cognitions, while attenuating symptoms, through behavioral exposure.
 - Treatment often includes cognitive therapy, exposure, and coping skills training.
- Examples of Cognitive Behavioral Therapies
 - Prolonged Exposure (PE)
 - Cognitive Processing Therapy (CPT)
 - Eye Movement Desensitization and Reprocessing (EMDR)



Treatment - Prolong Exposure Therapy

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- Prolonged Exposure therapy
 - Survivors repeatedly reexperience traumatic events via imaginal and in vivo exposure in safe situations
 - Trauma can become reprocessed and less painful
 - Following imaginal exposure, the patient will process with clinician to establish the emotional response to the experience in this safe environment
 - Patients are educated about common reactions to trauma
 - Usually administered for 9-12 weeks in twice weekly 90 Min Sessions
- Efficacy - More than 23 Randomized Control Trials have shown clinically significant improvements



Treatment - Cognitive Processing Therapy

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- Cognitive Processing Therapy
 - Principally a cognitive therapy
 - Exposure is achieved through written accounts of the traumatic event
 - Investigates the feelings elicited by the writing experience
 - Trains patient to challenge Faulty beliefs
- Efficacy – CPT vs PE or control showed as effective in reducing symptoms



Treatment - EMDR

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- Eye Movement Desensitization Reprocessing
 - CBT that incorporates saccadic eye movements
 - The patient imagine a traumatic event, while focusing on their feelings and cognitions
 - The clinician moves two fingers in front of their eyes to induce saccadic eye movements, until anxiety abates, then the patient is instructed to produce a more adaptive thought about the Event.
- Efficacy – remains uncertain, but some organizations recommend it as a treatment option.



Treatments - Pharmacotherapies

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- For people with treatment resistant PTSD, especially, combinations of Therapy and Medications are indicated
- SSRIs (first-line)
 - Paroxetine and Sertraline
 - Some evidence that more resilient patients will respond quickly
- Other Options
 - SSRI Fluoxetine
 - SNRI (Venlafaxine)
 - Other Antidepressants (Mirtazapine, Trazadone)
 - Prazosin (Nightmares)



Alternative Treatments

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- **Complementary and Alternative Medicine (CAM)**
 - Widely used, outside the standard practice
 - Includes Yoga, Acupuncture, meditation, Tai Chi, music, art, dance, and equine therapy
 - Estimates of 25-50% of patients are implementing
 - Current studies underway, but no solid evidence currently
- **Mindfulness Based Stress Reduction (MBSR)**
 - 9 weeks, group practice sessions
 - Teaches attending to the present moment in a non-judgmental, accepting way
 - Recent study showed improved subjective experience



Prognosis/Course

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- PTSD is often Chronic
 - 1/3 of patients recover in one year
 - 1/3 are symptomatic after ten years
- Established Relationship between Physical Health and Trauma
 - Evidence that development of PTSD may be the link
- Theories that neurochemical changes occur in PTSD
 - May have biological effect, for example increased HTN and Atherosclerotic Disease
- Mortality – Follow-up to NVVRS
 - From initial to follow-up 30 years later, Vietnam Veterans with PTSD were 2x as likely to have died.



Research and Novel Treatment

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- Evidence-based trials
 - Three elements: Best evidence-based research, clinical judgment, and patient preference
 - Studies have shown PE and CPT – the gold standard of treatment are not useful for all patients
- New studies looking at Transcranial Magnetic stimulation, Virtual Reality Exposure Therapy, Deep Brain Stimulation
- MDMA, Ketamine, Psilocybin, CBD with assisted exposure/psychotherapy

Source: Steenkamp, MM. True evidence based care for Posttraumatic Stress Disorder in the military. JAMA Psychiatry. 73(5):431-432, May 2016.

Summary

- A majority of people will experience trauma over their lifetime.
- There can be serious social, emotional, occupational, and health implications to the development of PTSD.
- Key to understanding PTSD is identifying patients through screening and treating them appropriately.
- A wide body of research exists and is ongoing. Key to the future is identifying what makes a person vulnerable or resilient.